Aniruddh Venkatesan

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Education

University of California, Berkeley *Aug. 2023–Dec 2025*
BA Mathematics with Honors, B.A Physics.

Research Experience

UC Berkeley SURF *May 2025–August 2025*
Examined proper varieties over the Fargues-Fontaine curve, proved GAGA result for vector bundles on proper varieties the Fargues-Fontaine curve. Advisor: Zeyu Liu.

UC Berkeley Math 274 *Jan 2025–June 2025*
Proved homological stability for completed cohomology of symplectic groups. Joint with Mingyang Cen. Advisor: Tony Feng

Experience

Reader/TA *Aug 2025 – Present*
raded problem sets for Math 256A: Graduate Algebraic Geometry I, Math 256B: Graduate Algebraic Geometry II, and Math 254B: Graduate Algebraic Number Theory II.

Papers

Interests: Algebraic number theory and algebraic geometry.

The Hodge-Tate Decomposition via Perfectoid Spaces *Senior Thesis*
Expository Thesis on the Hodge-Tate Decomposition for Rigid Spaces. Advisor: Sug Woo Shin

$\mathbf{FI}$-Modules and Homological Stability
Used $\mathbf{FI}$-modules to generate homological stability results of Calegari-Emerton from GL_n to Sp_{2n}.

GAGA Theorems for Varieties over the Fargues-Fontaine curve
Proved a GAGA result for vector bundles on proper varieties over the Fargues-Fontaine curve.

Arithmetic Volumes of Shtukas: Arbitrary Coweights
Generalized work of Wei-Wang to prove an arithemtic volume formula for all quasi-split semisimple groups, allowing all coweights.

Talks

The Local Langlands Correspondence and ℓ-adic representations *April 2025*
UC Berkeley Student Automorphic Forms Seminar

Completed Homological Stability For Sp_{2n} *May 2025*

Derived Categories and Algebraic Curves *December 2025*

p-adic Hodge Theory and Hodge-Tate Decompositions *December 2025*
UC Berkeley Student Number Theory Seminar

The Gross-Zagier Formula I: Number Fields *February 2026*
Student Number Theory Seminar

The Gross-Zagier Formula II: Function Fields *February 2026*
Student Number Theory Seminar

Conferences Attended

Joint Meetings in Mathematics *Jan. 2025*

Joint Meetings in Mathematics *Jan. 2026*

DL4Sci Deep Learning For Science 2026 Summer School *July 2026*

Outreach

Organizer *April. 2025–Present*

Undergraduate Seminar in Mathematics (SUM)

Organized weekly hour-long talks given by UC Berkeley undergraduates, on topics accessisble to undergraduates not typically covered in standard courses.

Teaching

Facilitator *Fall 2024, Spring 2025, Fall 2025*

MATH 174

MATH 174 is a student-lead course discussing topics in category theory: categories, functors, natural transformations, Yoneda Lemma, universal properties, (co)limits, adjunctions. As facilitator, wrote course notes, wrote and graded homework problems, and lectured (3 hrs/week).

Peer Tutor *Mar. 2022–June 2023*

De Anza College Student Learning Center

Worked one-on-one in drop-in environment for lower-division math as needed. Also co-wrote, organized, and delivered lecture-style content reviews for final exam preparation.

Mathematics Coursework

Course numbers beginning with 2 are graduate coursework. A * signals currently taking.

- 128A: Numerical Analysis
- H104: Real Analysis (Honors)
- H185: Complex Analysis (Honors)
- 215A: Algebraic Topology
- 214: Smooth Manifolds
- 242: Sympletic Geometry
- 198: Category Theory (student-led)
- 250B: Commutative Algebra
- 199: Perfectoid Spaces (reading course)
- 199 Higher Theta Series (reading course)
- 113: Abstract Algebra
- 126: Introduction to Partial Differential Equations
- 143: Elementary Algebraic Topology
- 254A: Algebraic Number Theory I
- 256A: Algebraic Geometry I
- 256B: Algebraic Geometry II
- 261B: Infinite Dimensional Lie Algebras
- 274: Topics in Algebra: Topics in Arithmetic Geometry
- 270: Advanced Topics in Mathematics: Survey of Deep Learning for Mathematicians
- 255: Algebraic Curves
- 196: Senior Thesis